

# Executive Summary

Homelab Security Monitoring Project

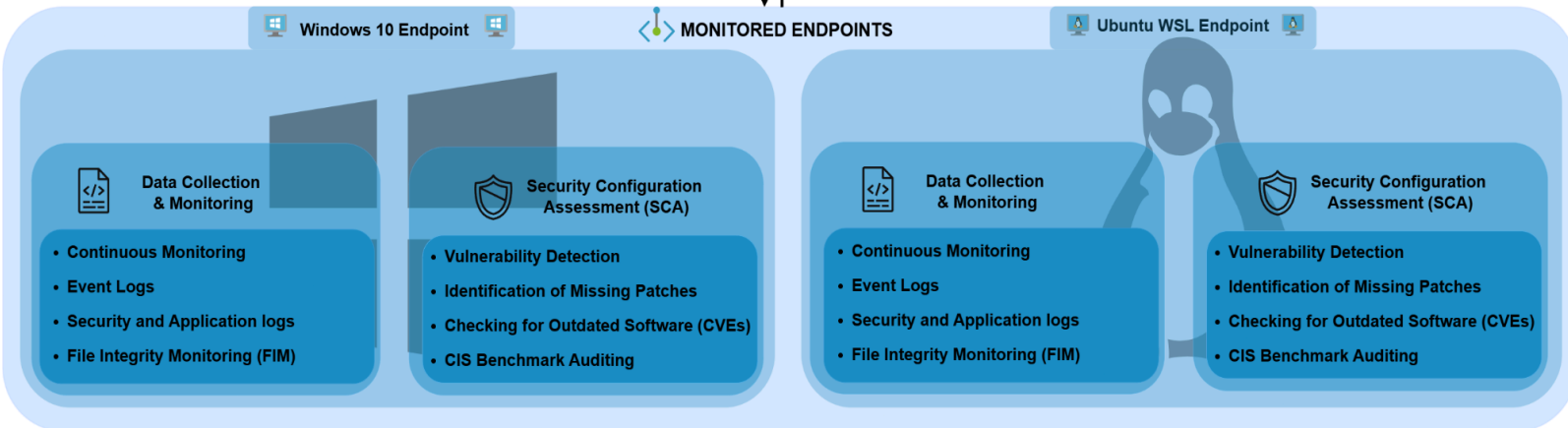
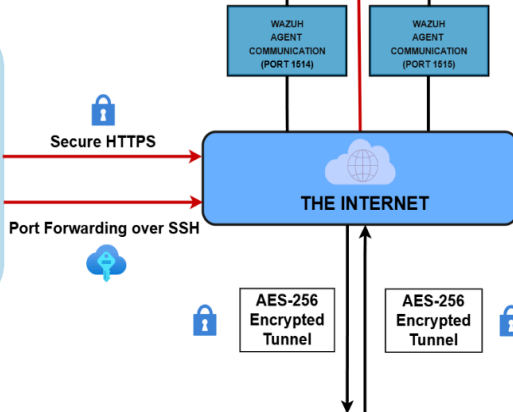
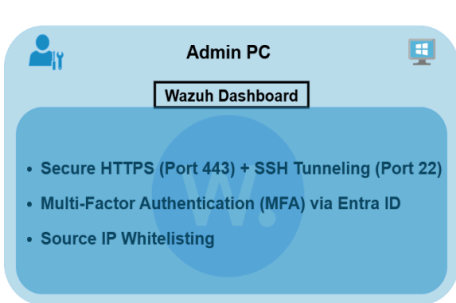
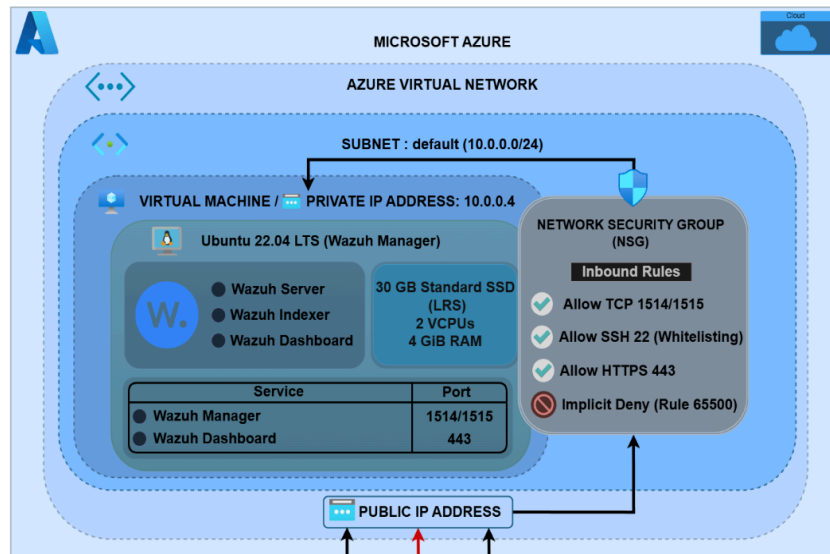
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**Date: March 2026**

**Platform: Wazuh SIEM (Microsoft Azure)**

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## Monitoring Homelab - Wazuh SIEM for Log Analysis and Threat Detection



## Objective

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To design and deploy a scalable, cloud-native **Security Information and Event Management (SIEM)** architecture that bridges the gap between decentralized endpoints and a centralized management system. This project was developed to simulate the real-world regulatory requirements and to help me implement governance, risk management and compliance techniques.

## Architecture

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The core of the environment consists of a **Wazuh Manager** hosted on an **Azure Virtual Machine (Ubuntu 22.04 LTS)**. This centralized hub orchestrates security telemetry from diverse endpoints. This includes a local **Windows 10** host and a **Windows Subsystem for Linux (WSL)** instance. Communication is secured via encrypted **Wazuh Agent** telemetry and managed through refined **Azure Network Security Group (NSG)** rules to ensure authentication and authorization.

## Governance & Compliance Alignment

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Beyond technical deployment, this lab serves as a functional proof-of-concept for the following compliance controls:

- **Continuous Monitoring (CA-7 / ISO 27001 A.12.4.1):** Real time visibility into unauthorized access attempts and system anomalies.
- **File Integrity Monitoring (FIM):** Tracking unauthorized changes to critical system files to ensure system Integrity and Availability.

The implementation further facilitates a comprehensive mapping of these security controls to widely recognized and relied upon industry frameworks:

- **NIST SP 800-53 (Revision 5):** Utilizing Wazuh's alerting capabilities to address the *Audit and Accountability (AU)* and *System and Information Integrity (SI)* control families, specifically providing the telemetry required for **continuous monitoring** and **incident response**.
- **ISO/IEC 27001:2022:** Supporting **Annex A Control 8.15 (Logging)** and **8.16 (Monitoring Activities)** by centralizing logs from Windows 10 and WSL endpoints into the Azure-hosted manager for analysis and retention.
- **Loi n° 05-20:** Directly aligns with **Article 7** which mandates the implementation of appropriate **monitoring and detection** mechanisms.